Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/22/2015
PI Name:	Liber, Howard Ph.D.		
Project Title:	Radiation quality and the relationship between induced telomere dysfunction and mutagenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ft. Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	11/01/2009	**End Date:**	10/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2014, per PI and NSSC information (Ed., 9/2/13) NOTE: Extended to 10/31/2013, per NSSC information (Ed., 12/18/12) NOTE: to be extended, per PI (Ed., 8/7/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bailey, Susan (Colorado State University)		
Grant/Contract No.:	NNX10AB36G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Radiation quality and the relationship between induced telomere dysfunction and mutagenesis. This research project is a collaboration between Drs. Howard L. Liber and Susan M. Bailey, both at Colorado State University (CSU) in Fort Collins, Colorado. The work specifically addresses Radiation Carcinogenesis Gap Cancer 3, which is: How can models of cancer risk be applied to reduce the uncertainties in radiation quality effects from solar particle events and galactic cosmic rays? The project explores the effects of radiation quality on the dose-response relationships for induction of telomere aberrations and gene mutations. Ultimately, it should relate these two potential cancer development processes by determining whether telomere-double strand break fusions are a significant contributor to the overall mutagenic response of a cell to ionizing radiation. Also, it explores various aspects of radiation-induced non-targeted (bystander) effects; novel aspects of the work include determining whether different radiation qualities vary in their abilities to induce bystander effects, and whether telomere-associated aberrations are induced as non-targeted responses. Telomere dysfunction has become an intensive subject for study and telomere-driven aberrations are considered important contributors to genomic instability. We are investigating two important aspects of this relationship. First, there has not yet been a detailed study of how different radiation qualities induce aberrations involving telomeres. Second, it is not yet established whether telomere-double strand break fusions are a significant contributor to viable chromosome alterations that actually produce the mutational events capable of driving carcinogenesis. Therefore, the goals are (i) to understand how radiation quality affects the induction of telomere aberrations and gene mutations, (ii) to determine whether a link may exist between telomere-double strand break fusions and mutagenesis, and (iii) to explore whether these radiations induce non-targeted effects. The primary goal of the bystander experiments is simply to ask whether radiation of different qualities induces non-targeted mutagenesis and telomere aberrations. Basically, we expect in each case to see either a yes or no response, although it is possible that we might see different magnitudes of response. The secondary goal is to determine the relative contributions of direct and non-targeted effects on the overall response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Dissecting the relationship between direct (targeted) and indirect (non-targeted or bystander) genetic effects of ionizing radiation will have significance for understanding the overall risks of both environmental and medical exposures to radiation. Determining whether telomere aberrations induced by ionizing radiation are found in cells surviving the exposures will have significance in understanding radiation-induced cancer.
Task Progress:	This research project was a collaboration between Drs. Howard L. Liber and Susan M. Bailey, both at Colorado State University in Fort Collins, Colorado. The work specifically addressed Radiation Carcinogenesis Gap Cancer 3, which is: How can models of cancer risk be applied to reduce the uncertainties in radiation quality effects from solar particle events and galactic cosmic rays? This research project explored the effects of radiation quality on the dose-response relationships for telomere aberrations and gene mutations. It was designed to relate these two potential cancer development processes by determining whether telomere-double strand break fusions are a significant contributor to the overall mutagenic response of a cell to ionizing radiation. Finally, it explored various aspects of radiation-induced non-targeted (bystander)effects; novel aspects of the work included determining whether different radiation qualities vary in their abilities to induce bystander signals, and whether telomere-associated aberrations can be induced as a non-targeted response. Telomere dysfunction has become an intensive subject of study and telomere-driven aberrations are considered important contributors to genomic instability. We investigated two important aspects of this relationship. First, there had not yet been a detailed study of how different radiation qualities induce aberrations involving telomeres. Second, it had not yet been established whether telomere-double strand break fusions are a significant contributor to viable chromosome alterations that actually produce the mutational events capable of driving carcinogenesis. Therefore, the goals were to (i) better understand how radiation quality affects the induction of telomere aberrations and gene mutations, (ii) to determine whether a link may exist between telomere-double strand break fusions and mutagenesis, and (iii) to explore whether these radiations induce non-targeted effects. 1. SPECIFIC AIMS Aim 1 was to produce detailed dose-response curves for mutagenesis and telomere-aberrations including telomere-double strand break fusions. Different radiations of varying energy and LET were examined, at doses of 0-100 cGy. These studies were done in a human lymphoblastoid cell model, which allowed examination of mutagenesis at the heterozygous thymidine kinase (TK) locus, as well as telomere aberrations. The experiments were done under conditions where the telomeres were either stable (capped) or unstable (uncapped, accomplished by reducing the levels of the telomere protection protein DNA-PKcs). Aim 2 was to extend the study of radiation quality on mutagenesis and telomere aberrations to the realm of non-targeted effects. Several questions were addressed, Including: a. Is non-targeted mutagenesis, known to be induced by low LET radiation, also radiation-quality dependent? How do non-targeted responses affect the overall shape of the dose-response curves for mutagenesis, for different radiation qualities? b. Are telomere-aberrations, including fusion with double strand breaks (DSBs), induced by bystander signals? Is there dependence on radiation-quality? c. There is a question whether radiation-induced double strand breaks trigger the production of bystander signals. Does a single double strand break in a fraction of cells produce a measurable bystander signal? d. As an extension of (c), if a single double strand break can generate a bystander signal, and a dysfunctional telomere can act like a double strand break, does a dysfunctional telomere produce bystander signal?

	Aim 3 was to determine if telomeres join with double strand breaks to produce viable mutants. Therefore this aim was designed to link telomere-dysfunction with the production of the types of mutations that can drive carcinogenesis. These experiments made use of a lymphoblast cell derivative that has a yeast I-Sce1 restriction site engineered into an intron of the active TK allele. Therefore introduction of the I-Sce1 enzyme produces a double strand break in that one site in the genome and nowhere else (fortuitously the18 base pair I-Sce1 sites do not occur randomly in the human genome). As in aim 1, the experiments were done under conditions where the telomeres are stable or unstable. Mutants that have lost TK function were analyzed by fluorescence in situ hybridization to determine the fraction that has joined TK sequence with telomere sequence.
Bibliography Type:	Description: (Last Updated: 01/22/2015)
Articles in Peer-reviewed Journals	Liber HL, Idate R, Warner C, Bailey SM. "Radiation quality and mutagenesis in human lymphoblastoid cells." Radiation Research. 2014 Oct;182(4):390-5. http://dx.doi.org/10.1667/RR13817.1 ; PubMed PMID: 25184374 , Oct-2014